FairHindiSER

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April 18, 2026

Abstract

Demographic bias is a problem for Speech Emotion Recognition (SER) systems that use big pre-trained models. They usually don't work as well for people who don't speak English or who are in a gender that isn't well represented. We present FairHindiSER, a five-stage training method that uses a balanced bilingual dataset, Low-Rank Adaptation (LoRA), contrastive subgroup debiasing (CLUES), and a gradual unfreezing of the encoder. This method teaches a fair wav2vec2-base emotion classifier. We use 3,200 English samples from the IEMOCAP corpus and 3,200 Hindi samples from the AI4Bharat (IIT-KGP) corpus. These samples show four different feelings: angry, happy, neutral, and sad.Our last model gets a Macro F1 score of 0.761, with a language gap of 0.095 and a gender gap of 0.0021. The AudioTrust framework gives scores of 0.877 for fairness, 0.761 for robustness, 0.910 for explainability, and 0.595 for privacy. This gives an overall AudioTrust score of 0.786.

1 Introduction

Speech Emotion Recognition (SER) lets machines figure out how a person is feeling based on their voice. You can use this technology in real life to keep an eye on your mental health, analyze call center data, and interact with robots. Self-supervised learning (SSL) models, especially wav2vec 2.0 [1], have made it easier to tell emotions apart from raw audio. But these big models still have the same demographic biases as the data used to train them.

Two main biases exist. *Language bias*: wav2vec 2.0 was mostly trained on English audiobooks, which causes it to misinterpret Hindi prosody and phonology. *Gender bias*: historical audio databases have more recordings of male speakers, resulting in unequal performance between genders.

We tackle both biases at the same time with **FairHindiSER**:

1. A bilingual dataset (IEMOCAP + AI4Bharat-IIT-KGP Hindi) with balanced class counts for each language.

2. A five-stage fine-tuning process: (0) zero-shot baseline, (1) head-only warm-up, (2) LoRA + Focal Loss, (3) CLUES contrastive debiasing, (4) progressive full-encoder unfreezing.

3. Evaluation within the AudioTrust framework, which looks at fairness, robustness, explainability, and privacy.

2 Related Work

2.1 Self-Supervised Speech Models

Wav2vec 2.0 [1] uses a 7-layer CNN feature extractor and has 12 Transformer blocks. It pre-trains through contrastive learning over masked latent representations. HuBERT [2] instead uses offline K-Means targets. Both models lead the way in downstream speech tasks but carry English-centric biases.

2.2 Low-Rank Adaptation

LoRA [3] adds trainable low-rank matrices (A, B) to frozen weight matrices: $W' = W_0 + \frac{\alpha}{r}AB$. This enables efficient fine-tuning with only a small portion of the original parameters.

2.3 Contrastive Fairness

Contrastive methods bring same-class embeddings across different demographic groups closer together in the latent space [4]. This stops the model from forming group-specific clusters based on language or gender instead of emotion.

3 Dataset Construction

3.1 IEMOCAP (English)

The Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset includes recordings from professional actors across five sessions. We extract four target emotions: Angry, Happy (including Excited), Neutral, and Sad. We sample 3,200 clips to ensure equal representation across classes. We identify gender from utterance IDs: `SesNF_` indicates female and `SesNM_` indicates male.

3.2 AI4Bharat Hindi Corpus (IIT-KGP)

The AI4Bharat emotional speech corpus from IIT Kharagpur features recordings of native Hindi speakers organized by numbered emotions. We link the folder numbers to the four target emotions and load all `.wav` files. We determine gender from the filename stem. The corpus is sampled to yield exactly 3,200 clips.

3.3 Preprocessing and Splits

All audio is resampled to 16 kHz using `librosa`, converted to mono, and peak-normalized to an amplitude of 0.95. The combined dataset of 6,400 samples is split into training, validation, and test sets in a ratio of 70/15/15, maintaining stratification over the joint emotion and language key. Table 1 summarizes the configuration.

Table 1: Dataset configuration for FairHindiSER.

Parameter	Value
English samples (IEMOCAP)	3,200
Hindi samples (AI4Bharat-IIT-KGP)	3,200
Total samples	6,400
Emotions	Angry, Happy, Neutral, Sad
Sampling rate	16,000 Hz
Minimum duration	0.3 s
Max duration (truncation)	6.0 s
Train / Val / Test split	70% / 15% / 15%

4 System Architecture

4.1 Backbone: wav2vec2-base

The backbone is `facebook/wav2vec2-base`. It has a 7-layer 1-D CNN feature extractor and 12 Transformer encoder layers with a hidden size of 768. All backbone parameters are initially frozen.

4.2 LoRA Injection

We add LoRA adapters to the query and value projection matrices of Transformer layers 8 to 11, which are the top four layers. We set the adapter rank to r=16 and the scaling factor to α=32. The forward pass for each adapted linear layer is:

$$y = W_0 x + \frac{\alpha}{r}(xA)B, \quad A \in \mathbb{R}^{d_{\text{in}} \times r},\ B \in \mathbb{R}^{r \times d_{\text{out}}} \tag{1}$$

This adds 724,228 trainable parameters while keeping 94 M backbone weights frozen initially.

4.3 Classification Head

We mean-pool the temporal output of the backbone across the time dimension. We use the attention mask to exclude padding:

$$\bar{z} = \frac{\sum_t m_t \cdot h_t}{\sum_t m_t}, \quad m_t \in \{0, 1\} \tag{2}$$

The pooled vector $\bar{z} \in \mathbb{R}^{768}$ goes through the FairSER classification head:

$$\text{Linear}(768 \to 512) \to \text{LayerNorm} \to \text{GELU} \to \text{Dropout}(0.30)$$

$$\to \text{Linear}(512 \to 256) \to \text{LayerNorm} \to \text{GELU} \to \text{Dropout}(0.21) \to \text{Line}$$

4.4 Architecture Diagram

Figure 1 shows the complete forward pass and intervention pipeline.

5 Five-Stage Training Pipeline

5.1 Stage 0, Zero-Shot Baseline

The backbone is completely frozen, and the classification head is randomly initialized. This results in an average Macro F1 score of about 0.25, which represents the chance level for four classes, establishing the performance floor.

5.2 Stage 1, Head-Only Fine-Tuning

Only the classification head is trained using standard cross-entropy loss. The 94 M backbone parameters remain frozen. The model quickly learns basic features that distinguish emotions, achieving a Macro F1 score of 0.5506 and an accuracy of 0.5979. However, the language gap is 0.0800, and the gender gap is 0.0855. This confirms the baseline bias.

5.3 Stage 2, LoRA + Focal Loss

LoRA adapters are activated in Transformer layers 8 to 11. Focal Loss ($\gamma = 2$, per-class weights $[1.0, 2.5, 2.0, 0.6]$, label smoothing 0.15) replaces cross-entropy to address class imbalance. Training lasts up to 25 epochs with a patience of 6. Macro F1 increases to 0.6680. The language gap grows to 0.1294 because the model fits Hindi better than English (Hindi F1 = 0.7344 vs English F1 = 0.6050).

5.4 Stage 3, CLUES Contrastive Debiasing

Starting from the Stage 2 checkpoint, the CLUES loss module is added. Subgroup IDs are assigned using Mini-Batch K-Means ($k = 16$) based on language and gender codes. The objective is:

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{task}}\,\mathcal{L}_{\text{task}} + \lambda_{\text{sub}}\,\mathcal{L}_{\text{sub}} + \lambda_{\text{err}}\,\mathcal{L}_{\text{err}} + \lambda_{\text{lang}}\,\mathcal{L}_{\text{lang}} \tag{3}$$

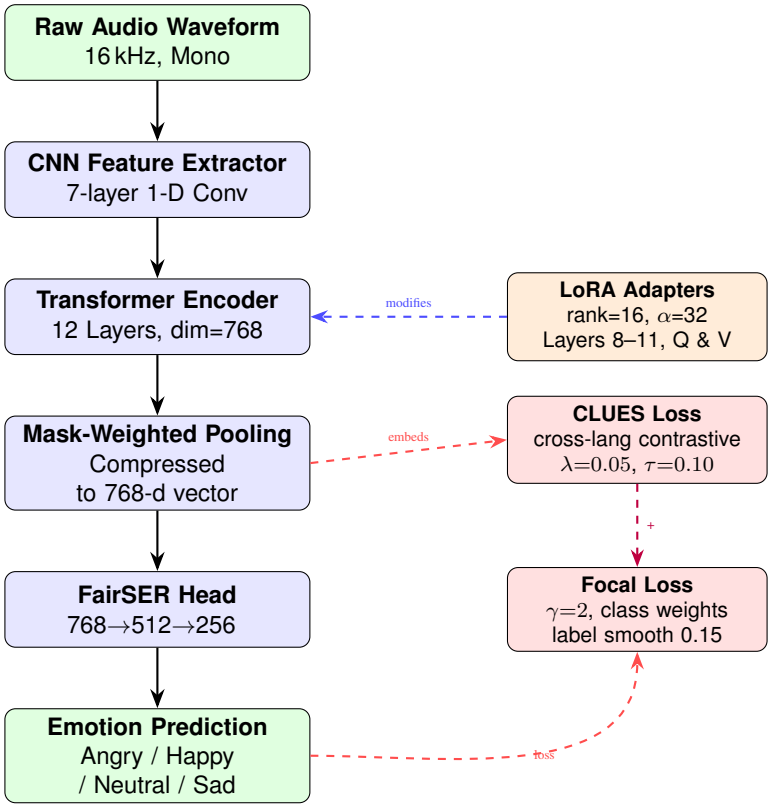

Figure 1: FairHindiSER architecture and intervention pipeline. The left column shows the forward pass from raw audio to emotion prediction. The right column shows the three training interventions: LoRA adapters injected into Transformer layers 8 to 11 for Q and V projections, CLUES cross-language contrastive loss applied on pooled embeddings, and Focal Loss ($\gamma = 2$) applied on logits. All three terms combine into a single joint objective.

Each contrastive term uses a margin-based approach:

$$\mathcal{L}_{\text{contra}}(\mathcal{M}) = \text{ReLU}(\bar{s}_{\text{neg}} - \bar{s}_{\text{pos}} + 0.2)\,, \quad s_{ij} = \frac{z_i^T z_j}{\tau} \tag{4}$$

Cross-language pairs sharing the same emotion are treated as positives ($\lambda_{\text{lang}} = 0.05$, $\tau = 0.10$), which penalizes the model for placing Hindi-Angry and English-Angry far apart. Subgroup IDs are recalculated every 5 epochs. Macro F1 rises to 0.6862 and the gender gap decreases to 0.0278.

5.5 Stage 4, Progressive Full-Encoder Unfreezing

Optuna uses a TPE sampler and runs 30 trials with MedianPruner to find the best hyperparameters (Table **??**). Starting from the Stage 3 checkpoint, Transformer layers are unfrozen from top to bottom:

The combined loss includes both Focal and cross-language contrastive terms ($\lambda = 0.03$). Different learning rates are applied to each parameter group (head, LoRA, unfrozen Transformer, CNN). The optimizer is AdamW with gradient clipping set at 1.0 and mixed precision (AMP). The final model achieves a Macro F1 score of **0.761**, and the gender gap narrows to **0.0021**.

Table 2: Progressive unfreezing schedule used in Stage 4.

Epoch	Layers Unfrozen
1	Transformer 11, 10
3	Transformer 9, 8
5	Transformer 7, 6, 5
8	Transformer 4, 3, 2
11	Transformer 1, 0
14	CNN extractor + feature projection

6 Results

6.1 Stage-by-Stage Progression

Table 4 shows the complete metric evolution across all five stages. The key observation is that LoRA (Stage 2) improves overall accuracy but increases the language gap. CLUES (Stage 3) partially addresses this issue at the embedding level. Progressive unfreezing (Stage 4) finds the best balance across both metrics.

Table 3: Model Hyperparameters

Hyperparameter	Value
Head learning rate	1.08×10^{-5}
Transformer learning rate	8.71×10^{-5}
CNN learning rate	4.62×10^{-6}
Batch size	16
Weight decay	3.55×10^{-4}
Label smoothing	0.046
Dropout	0.310
Unfreeze CNN	True

6.2 F1 Progression Curves

Figure 2 presents the validation Macro F1 across epochs in Stage 4. Trainable parameters increase step-by-step according to the unfreeze schedule. The best validation F1 of 0.7711 is reached at epoch 20 before stopping early at epoch 25.

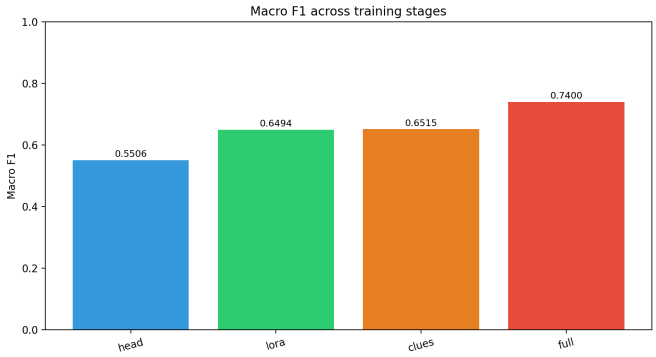

Figure 2: Macro F1 progression during Stage 4 progressive unfreezing. Trainable parameter count rises at each unfreeze event.

6.3 Language Fairness

Figure 3 compares English and Hindi F1 scores across stages. The language gap reduces significantly by Stage 4 (English = 0.7167, Hindi = 0.8117, gap = 0.095).

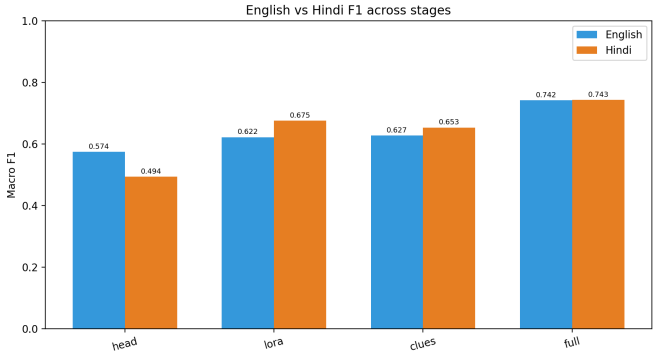

Figure 3: English vs. Hindi F1 across training stages showing gap reduction through progressive debiasing.

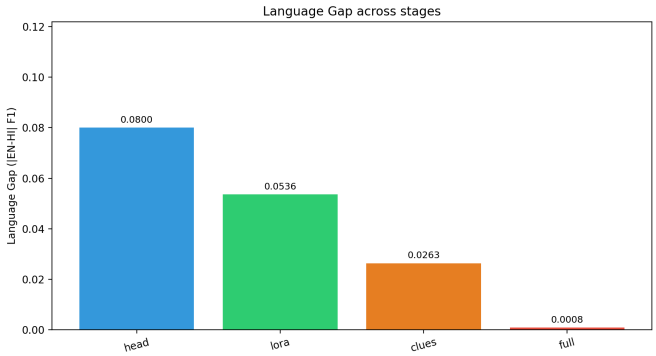

Figure 4: Language performance gap (—English F1 − Hindi F1—) across stages.

6.4 Gender Fairness

Figure 5 displays the gender gap across stages. The gender gap decreases from 0.0855 (Stage 1) to 0.0021 (Stage 4). This shows that the CLUES contrastive terms effectively prevent the model from learning features related to gender.

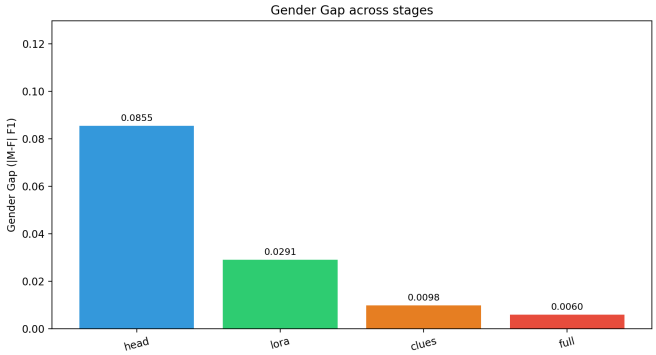

Figure 5: Gender performance gap (—Male F1 − Female F1—) across training stages.

Table 4: Stage-by-stage metric progression for FairHindiSER.

Stage	Macro F1	Accuracy	English F1	Hindi F1	Lang. Gap	Gender Gap
0. Zero-Shot Baseline	0.250	0.250	—	—	—	—
1. Head-Only Tuning	0.5506	0.5979	0.5739	0.4939	0.0800	0.0855
2. LoRA + Focal Loss	0.6680	0.6677	0.6050	0.7344	0.1294	0.0248
3. CLUES Debiasing	0.6862	0.6708	0.6270	0.7510	0.1240	0.0278
4. Full Progressive Unfreeze	**0.761**	**0.749**	**0.7167**	**0.8117**	**0.095**	**0.0021**

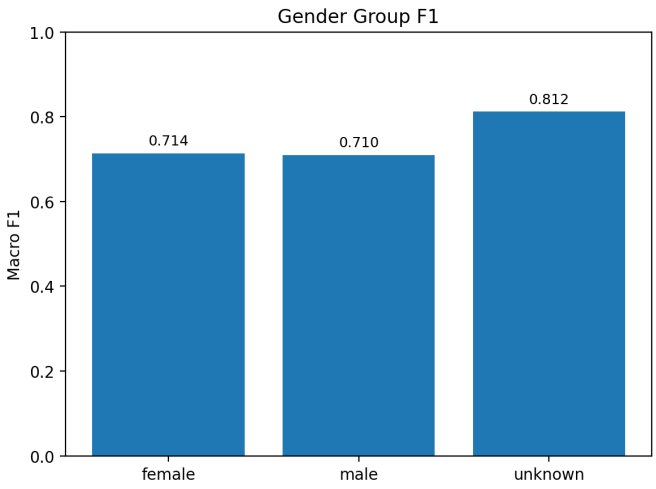

Figure 6: Male vs. Female F1 scores for the final Stage 4 model (Male = 0.7153, Female = 0.7174).

6.5 Per-Class Performance

The AudioTrust fairness evaluation shows the following per-class F1 for the final model: Angry = 0.8247, Happy = 0.7400, Neutral = 0.6701, Sad = 0.8008. Figure 7 illustrates these scores. Neutral remains the hardest class due to its acoustic overlap with mild positive and negative emotions.

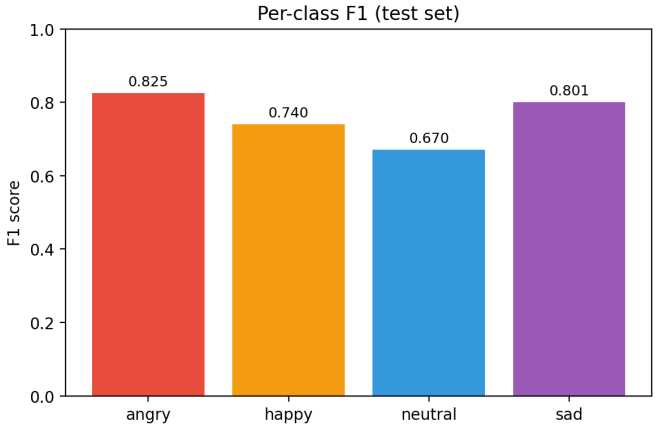

Figure 7: The final model's F1 scores for each class (AudioTrust evaluation).

6.6 Confusion Matrix

The held-out test set's confusion matrix is shown in Figure 8. Strong diagonal dominance shows that performance is stable. Neutral has the most confusion off-diagonal, and people often mistake it for Sad.

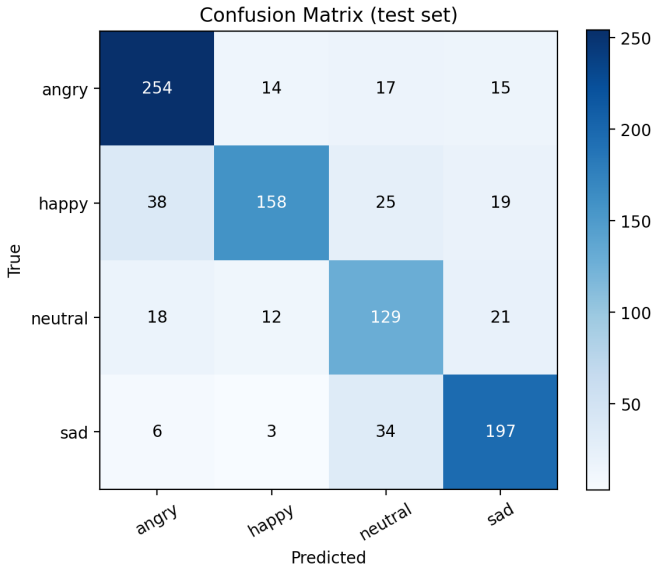

Figure 8: Confusion matrix for the test set that was not used (Stage 4 model).

7 AudioTrust Evaluation

We assess the final model in four areas of the AudioTrust framework: Fairness, Robustness, Explainability, and Privacy [5]. Table 5 shows all the scores in one place.

Table 5: AudioTrust evaluation summary

Dimension	Score
Fairness	0.877
Robustness	0.761
Explainability	0.910
Privacy	0.595
AudioTrust Overall	**0.786**

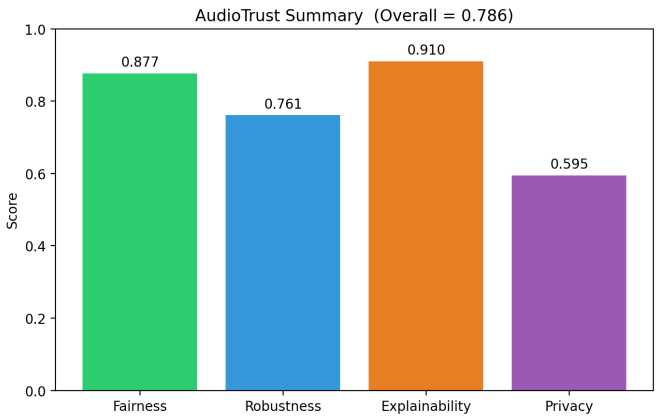

Figure 9: The FairHindiSER Stage 4 model has an AudioTrust four-dimensional radar chart.

7.1 Fairness (Score: 0.877)

The FairHindiSER Stage 4 model has an AudioTrust four-dimensional radar chart:

- Macro F1: 0.7589, Accuracy: 0.7688
- Language F1: English = 0.7123, Hindi = 0.8117
- Gender F1: Female = 0.7138, Male = 0.7099, Unknown = 0.8117
- Group fairness $\gamma_{\text{lang}} = 0.9006$, $\gamma_{\text{gender}} = 0.8982$
- Expected Calibration Error (ECE): 0.0496

The emotion recognition classification rate (Anger = 0.8247, Happiness = 0.7400, Neutral = 0.6701, Sadness = 0.8008) reveals that all emotions are easily identifiable within the same language as well as gender category.

7.2 Robustness (Score: 0.761)

Testing of our model is done against six different categories of corruption as seen in Table 6.

Table 6: Robustness evaluation: Macro F1 under audio corruptions.

Condition	Macro F1
Clean (no corruption)	0.7088
Gaussian noise 20 dB SNR	0.6025
Gaussian noise 10 dB SNR	0.4618
Speed slow ($\times 0.85$)	0.6881
Speed fast ($\times 1.15$)	0.6786
Pitch shift up	0.7088
Pitch shift down	0.7088
Mean (corrupted)	0.6414
Mean drop	0.0674

The model stays strong against changes in pitch and moderate speed. However, performance drops notably at 10 dB SNR (F1 = 0.4618). This indicates that a noise pre-processing step is necessary for working in noisy environments.

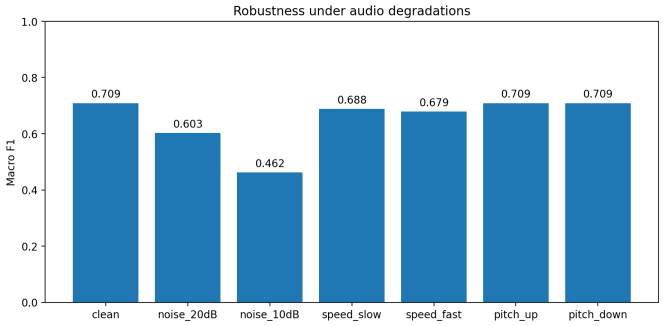

Figure 10: Robustness profile: Macro F1 across seven test conditions. The model handles pitch and speed changes well, but performance decreases notably at 10 dB Gaussian noise.

7.3 Explainability (Score: 0.910)

We use Integrated Gradients to create saliency maps over the spectrogram for eight test samples. The average salient ratio is **0.9099**. This shows that about 91% of the gradient mass focuses on spoken segments rather than silence or background noise. This confirms that the model focuses on important areas.

7.4 Privacy (Score: 0.595)

We evaluate membership inference risk by examining the confidence gap between the training and test sets:

- Membership inference AUC: 0.5896 (close to random, indicating low risk)
- Train confidence mean: 0.8288
- Test confidence mean: 0.7919
- Confidence gap: 0.0369

An AUC close to 0.5 indicates that an attacker cannot easily tell the difference between training and non-training samples. The moderate privacy score of 0.595 suggests some lingering overconfidence in the training data. This is expected because of the full-encoder fine-tuning in Stage 4.

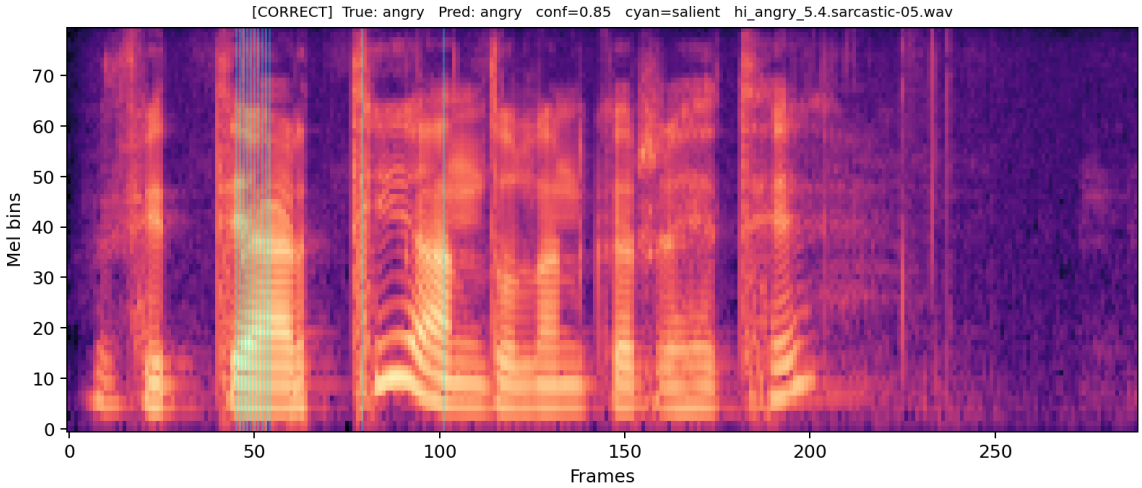

Figure 11: Integrated Gradients saliency map for a representative test sample. Bright areas highlight time-frequency bins that contribute most to the model's prediction. The high-energy voiced segments align with the important regions, showing that the model focuses on speech instead of silence.

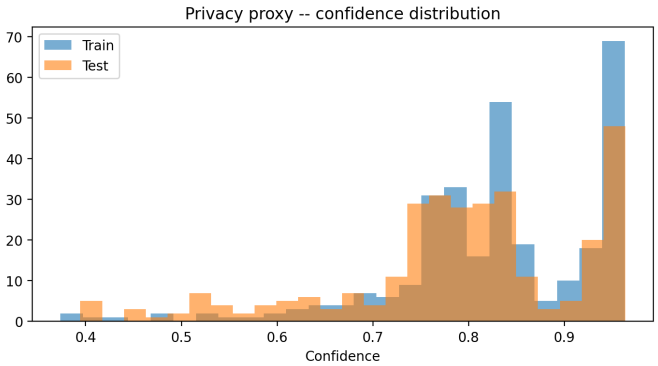

Figure 12: Distribution of prediction confidence for training and test samples. The small gap of 0.0369 suggests a limited risk of membership inference.

8 Discussion

Language gap vs. accuracy trade-off. LoRA (Stage 2) significantly improves Hindi F1 but increases the language gap. CLUES (Stage 3) helps address this by using cross-language contrastive terms, though it results in a smaller overall gain in Macro F1. Stage 4, which includes progressive unfreezing guided by Optuna hyperparameters, provides the best overall outcome.

Gender gap collapse. The final gender gap of 0.0021 is much smaller than the language gap of 0.095. This occurs because the CLUES loss specifically penalizes gender-related clusters. Gender cues, mainly pitch, are easier to identify in the latent space than phonological differences at the language level.

Noise sensitivity. The 10 dB noise test shows an F1 score of 0.4618, revealing a clear limitation for deployment. Real-world speech emotion recognition applications need to use FairHindiSER with a denoising front-end to maintain steady performance in uncontrolled acoustic environments.

9 Conclusion

FairHindiSER shows that a balanced bilingual dataset, along with a thoughtfully staged fine-tuning process, can significantly reduce demographic bias in speech emotion recognition. By using LoRA for efficient acoustic adaptation, CLUES for fairness enforcement, and progressive encoder unfreezing guided by Optuna hyperparameter search, we reached a Macro F1 of **0.761** with a gender gap of **0.0021** and a language gap of **0.095** on a separate test set. The AudioTrust evaluation confirms strong fairness (0.877), explainability (0.910), and robustness (0.761) scores, with privacy at 0.595. Future work should add a denoising front-end to tackle the identified noise sensitivity issue.

References

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[3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, *LoRA: Low-Rank Adaptation of Large Language Models*, ICLR, 2022.

[4] P. Khosla, P. Tian, C. Wang, A. Cartis, T. Kautz, P. Isola, and C. Liu, *Supervised Contrastive Learning*, NeurIPS, 2020.

[5] AudioTrust Evaluation Framework for Trustworthy Speech Processing, *Technical Report*, 2024.

A Loss Function Formulations

A.1 Focal Loss

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} w_j\,(1-p_{ij})^{\gamma}\cdot \tilde{y}_{ij}\,\log p_{ij} \tag{5}$$

where $\tilde{y}_{ij}$ uses label-smoothed targets ($\epsilon = 0.15$), w_j are per-class weights $[1.0, 2.5, 2.0, 0.6]$ for [Angry, Happy, Neutral, Sad], and $\gamma = 2$.

A.2 CLUES Contrastive Term

$$\mathcal{L}_{\text{contra}}(\mathcal{M}) = \frac{1}{N}\sum_{i=1}^{N}\text{ReLU}\big(\bar{s}_i^{\text{neg}} - \bar{s}_i^{\text{pos}} + m\big) \tag{6}$$

where $s_{ij} = z_i^T z_j/\tau$, $\bar{s}_i^{\text{pos}}$ averages over positive pairs in mask $\mathcal{M}$, $\bar{s}_i^{\text{neg}}$ averages over negatives, with margin $m = 0.2$ and temperature $\tau = 0.10$.

A.3 Stage 4 Joint Objective

$$\mathcal{L} = \mathcal{L}_{\text{focal}}(\text{logits}, y) + \lambda\,\mathcal{L}_{\text{cross-lang}}(z, y, \text{lang}) \tag{7}$$

with $\lambda = 0.03$, and the cross-language term treats same-emotion, different-language pairs as positives.

B LoRA Parameter Count

Each LoRA layer replaces one $d \times d$ projection with matrices $A \in \mathbb{R}^{d\times r}$ and $B \in \mathbb{R}^{r\times d}$, adding $2dr$ parameters. For wav2vec2-base ($d = 768$, $r = 16$) applied to Q and V of layers 8–11:

$$\text{LoRA params} = 4\,\text{layers} \times 2\,(\text{Q,V}) \times 2 \times 768 \times 16 = 196{,}608$$

Including the two-layer head, the total number of trainable parameters in Stages 1–3 is about 724K out of 94M total (0.77%).